

Three fermion families from the automorphism algebra of their own carrier: a conditional Killing-contact selection theorem

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Abstract

We establish three representation-theoretic statements about the family structure of chiral matter. (i) Among the complex irreducible representations of all simple compact Lie algebras, enumerated exhaustively at dimension ≤ 16 , the $\text{Spin}(10)$ half-spinor is the unique object whose restriction to the Standard-Model subalgebra reproduces exactly one family; its sixteenth state — a total Standard-Model singlet, ν^c — is thereby forced, not chosen. (ii) If the multiplicity space of families is required to carry, as its covariance algebra, the automorphism algebra of its own geometric carrier — generated by the geometry rather than imposed upon it — then the number of families is bounded a priori by three. An invariant nondegenerate bilinear form alone does not force equality: a one-dimensional abelian algebra has such forms although its Killing form vanishes. (iii) Under the additional, explicit requirement that the contact be the nondegenerate adjoint-trace (Killing) form, the carrier is the rational curve, the family space is its Möbius algebra $\mathfrak{sl}(2, \mathbb{C})$, and $N_{\text{fam}} = \dim \mathfrak{sl}(2, \mathbb{C}) = 3$. On this selected branch the invariant Majorana contact direction is unique up to scale and equals the Killing form, $B = 2\sqrt{3} K_{\text{tr}}$ with $K_{\text{tr}}^2 = \frac{1}{3}I$; the same simplicity that selects three families guarantees that the seesaw contact exists, is unique, and is invertible. No phenomenological claim is made. The three-family equality is conditional on the strengthened Killing-contact hypothesis; the bound $N_{\text{fam}} \leq 3$ does not require it. The finite enumeration and contact identities are machine-verified in exact arithmetic in a public repository.

I. INTRODUCTION

Why the chiral matter of the Standard Model comes in three identical families is an old structural question. Three traditions address it. The family-symmetry tradition imposes a horizontal group — continuous or discrete, A_4 , S_4 , $\text{SU}(2)$, $\text{SU}(3)$ and their relatives — on the multiplicity space and derives textures from its breaking [1]; the group is chosen, not derived. The index-theorem tradition, descending from compactified string theory, counts families as a topological index of an internal space [2]; the number reflects the choice of internal geometry. A third, division-algebra tradition relates the family triplication to exceptional algebraic structures (octonionic trialities and their relatives; see Ref. [3] for the

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algebraic setting of grand unification); there the family symmetry arises as a discrete outer automorphism of a division-algebra construction.

This Letter proves a different statement, which we have not found in the literature in this combination. We require the family multiplicity space to be *self-carried*: the symmetry acting on families is not imposed and not inherited from a larger construction, but is the automorphism algebra of the geometric object whose sections *are* the families. Under this requirement the family number is bounded a priori by three (Lemma 2). A correction is essential at the saturation step: invariance and nondegeneracy alone do not exclude the one-dimensional abelian branch. We therefore separate that weak contact hypothesis from the stronger requirement that the contact be the adjoint-trace (Killing) form and be nondegenerate. This explicit Killing-contact condition selects the unique case in which the bound is saturated (Theorem 2); and the resulting contact direction is then not merely *like* a Killing form but *is* one (Theorem 3). Combined with an exhaustive enumeration selecting the Spin(10) half-spinor as the unique single-object realization of one family (Theorem 1), the package gives a conditional answer to “why three” and “why a Killing seesaw direction” with one piece of standard mathematics used in a nonstandard place: simplicity of $\mathfrak{sl}(2)$.

All results are elementary as mathematics; the claim is the assembly. Every statement below is verified by standalone machine audits (exact rational arithmetic for the enumeration; see Sec. VII), and nothing here asserts a phenomenological fit: what the hypotheses do and do not deliver is stated explicitly in Sec. VI.

II. HYPOTHESES

Three base hypotheses and one explicit strengthening are used, stated here informally and used precisely in the theorems.

(H1) *Face projection with an observed-content anchor.* One family of chiral matter is realized as a single complex irreducible representation R of a simple compact group $G \supseteq G_{\text{SM}}$, whose restriction to $G_{\text{SM}} = \text{SU}(3) \times \text{SU}(2) \times \text{U}(1)_Y$ contains exactly the fifteen observed Weyl states of one family (with their multiplicities), plus possibly Standard-Model singlets, and no exotics; the gauge theory of R must be anomaly-free.

(H2) *Self-carried index.* The family multiplicity space is $V = H^0(\Sigma, \mathcal{L})$ for a holomorphic line bundle $\mathcal{L}$ on a compact Riemann surface Σ , and the covariance algebra acting on families

is required to be $\mathbf{aut}(\Sigma) = H^0(\Sigma, T_\Sigma)$ acting on V — with the self-carried clause $\mathcal{L} \simeq T_\Sigma$, so that the family space is the automorphism algebra itself (the adjoint module), not a module over an external symmetry. An existence anchor $N_{\text{fam}} \geq 1$ is assumed.

(H3) *Invariant contact.* The family algebra admits an invariant symmetric bilinear form, nondegenerate on V — the structure a family-covariant Majorana (seesaw) mass sector requires.

(H3⁺) *Killing contact.* The canonical contact is required to be the adjoint-trace form

$$B_{\text{ad}}(x, y) = \text{Tr}_{\mathbf{aut}(\Sigma)}(\text{ad}_x \text{ad}_y)$$

up to a nonzero normalization, and this form is required to be nondegenerate. Equivalently for the finite-dimensional complex Lie algebras occurring below, the family algebra is semisimple. This is an additional selection axiom, not a consequence of H3.

H1 is the standard grand-unification setting [4, 5] sharpened into an enumeration hypothesis; H2 replaces an imposed family group by a generated one; H3 is the minimal algebraic demand of a covariant seesaw. H3⁺ is the stronger demand needed for the Killing-contact selection. Neither H2 nor H3 alone singles out a specific curve.

III. ONE FAMILY: THE ENUMERATED HALF-SPINOR

Lemma 1 (Chirality forces complexity). *If $\text{Res}_{G_{\text{SM}}}^G R$ satisfies the observed-content anchor, R is a complex representation of G .*

Proof. Restriction commutes with conjugation, and the anchored content is not self-conjugate: its hypercharge multiset contains $Y = 1/6$ with multiplicity six and $Y = -1/6$ with multiplicity zero, a mismatch that extra self-conjugate singlets cannot repair. $\square$

Since a maximal torus of G_{SM} embeds into one of G , the rank of G is at least four. Only the A_n , D_{2k+1} , and E_6 series admit complex irreducibles (the conjugation involution $-w_0$ is computed from the Cartan matrix for every simple algebra of rank 4–15 and for E_6, E_7, E_8, F_4 in the machine audit; textbook context in Ref. [6]). The Weyl dimension formula is strictly increasing in each Dynkin label, so the following finite scan is exhaustive.

Proposition 1 (Enumeration at dimension ≤ 16). *Among all simple compact groups of rank ≥ 4 (every algebra of rank 4–15 scanned directly; rank ≥ 16 contributes nothing, the*

smallest rank-16 irreducibles having dimensions 17, 33, 32, 32), the complex irreducible representations of dimension 15 or 16 are: $\text{Sym}^2 5$ of $\text{SU}(5)$ and $\Lambda^2 6$ of $\text{SU}(6)$ (both of which fail the content anchor: every embedding produces a color sextet, octet, or a wrong antitriplet count); the fundamentals of $\text{SU}(15)$ and $\text{SU}(16)$ (gauge-anomalous: on an exact integer diagonal generator $\text{Tr} T^3 = 2730$ and 3360 [7, 8]); and the $\text{Spin}(10)$ half-spinor 16, together with conjugates.

Theorem 1 (Enumerated one-family uniqueness). *Under H1 the minimal-dimension realization exists, is unique, and is the $\text{Spin}(10)$ half-spinor, with*

$$\text{Res}_{G_{\text{SM}}}^{\text{Spin}(10)} 16 = Q \oplus L \oplus u^c \oplus d^c \oplus \nu^c \oplus e^c,$$

multiplicities (6, 2, 3, 3, 1, 1) as Weyl states. The half-spinor is anomaly-free: with the 16 built explicitly from five fermionic oscillators (chirality = occupation parity), $\text{Tr}(\{T_a, T_b\}T_c) = 0$ holds exactly over all 45^3 generator triples; the nonperturbative check is $\pi_4(\text{Spin}(10)) = 0$ [9].

Corollary 1 (The sixteenth state is a prediction). *No fifteen-dimensional single-object realization of one family exists; the sixteenth state — a total Standard-Model singlet, ν^c — is forced. The essential seesaw ingredient is an output of H1, not an input. (Instructive near-misses: $\mathfrak{so}(9)$ has a real spinor of dimension exactly 16 and $\mathfrak{so}(15)$ a real vector of dimension 15; both fail Lemma 1. Chirality, not size, selects $\text{Spin}(10)$.)*

On the surviving object $Y = T_{3R} + (B - L)/2$, and the weight-level traces give $\text{Tr} Y^2 = 10/3$, $\text{Tr} T_{3L}^2 = 2$, hence the canonical normalization $k_Y = 5/3$.

IV. THREE FAMILIES: THE SELF-CARRIED BOUND AND CONDITIONAL CARTAN SELECTION

Lemma 2 (Genus ladder). *For a compact Riemann surface Σ_g ,*

$$h^0(\Sigma_g, T_{\Sigma_g}) = \begin{cases} 3, & g = 0, \\ 1, & g = 1, \\ 0, & g \geq 2. \end{cases}$$

Consequently, under H2, $N_{\text{fam}} \leq 3$ a priori, with equality if and only if the family curve is rational.

Proof. $T_{\Sigma_g} = K_{\Sigma_g}^{-1}$ has degree $2 - 2g$. For $g = 0$, Riemann–Roch gives $h^0 - h^1 = 3$ and Serre duality gives $h^1 = h^0(K^2) = h^0(\mathcal{O}(-4)) = 0$; for $g = 1$ the tangent bundle is trivial; for $g \geq 2$ its degree is negative [10, 11]. $\square$

Lemma 3 (Conditional Cartan selection). *Among the cases $g \in \{0, 1\}$ allowed by the existence anchor, both admit nondegenerate invariant symmetric bilinear forms, but only $g = 0$ satisfies $H3^+$.*

Proof. For $g = 1$ the automorphism algebra is one-dimensional abelian; the Killing form of an abelian algebra vanishes identically. Nevertheless, if e is a basis, every form $B(e, e) = c$ with $c \neq 0$ is symmetric, invariant, and nondegenerate. Thus $H3$ does not eliminate this branch. It does fail $H3^+$ because its adjoint-trace form is zero. For $g = 0$, $\mathfrak{aut}(\mathbb{CP}^1) = \mathfrak{sl}(2, \mathbb{C})$ is simple, so its Killing form is nondegenerate by the Cartan criterion and $H3^+$ holds. $\square$

Theorem 2 (Conditional three-family selection). *Under $H2$ and $H3^+$: the family curve is $\Sigma \simeq \mathbb{CP}^1$; the family space is canonically the Möbius algebra $V = H^0(\mathbb{CP}^1, T_{\mathbb{CP}^1}) \simeq \mathfrak{sl}(2, \mathbb{C})$; the carrier is $T_{\mathbb{CP}^1} \simeq \mathcal{O}(2)$; and $N_{\text{fam}} = \dim \mathfrak{sl}(2, \mathbb{C}) = 3$. Under $H2$ alone, or under $H2$ and $H3$, the rigorously established conclusion is only $1 \leq N_{\text{fam}} \leq 3$.*

Two structural remarks. First, the zero divisor of any regular semisimple element $X_h \in V$ (any regular semisimple Möbius flow) is a sum of two distinct points with $\deg \text{div}(X_h) = c_1(T)[\mathbb{CP}^1] = \chi(S^2) = 2$: the “two centers” often postulated in family-geometry constructions are the two fixed points of a Cartan flow — Poincaré–Hopf, not an assumption. Second, the self-carried clause in $H2$ does real selective work: for $g = 0$ one has $h^0(K^{-m}) = 2m + 1$, so “some power of the anticanonical bundle” would allow any odd family number. Only for $m = 1$ is the family space itself the algebra generating the family covariance — the adjoint module — rather than a module over an external symmetry. This is precisely what separates the present mechanism from the imposed-symmetry tradition.

V. THE CONTACT IS THE KILLING FORM

A family-covariant Majorana sector requires an invariant symmetric bilinear form on V .

Lemma 4 (Uniqueness up to scale). *$\text{Sym}^2(\text{Sym}^2 \mathbb{C}^2) \simeq \text{Sym}^4 \mathbb{C}^2 \oplus \text{Sym}^0 \mathbb{C}^2$, so the invariant symmetric bilinear form on the adjoint (spin-1) module of $\mathfrak{sl}(2, \mathbb{C})$ is unique up to scale [12].*

Theorem 3 (Killing contact). *In the spherical orthonormal family basis (T_{+1}, T_0, T_{-1}) , $T_{\pm 1} = \mp J_{\pm}/\sqrt{2}$, $T_0 = J_z$ (spin-1 matrices), the Killing form of the family algebra is*

$$B = \begin{pmatrix} 0 & 0 & -2 \\ 0 & 2 & 0 \\ -2 & 0 & 0 \end{pmatrix} = 2\sqrt{3} K_{\text{tr}}, \quad K_{\text{tr}} = \frac{1}{\sqrt{3}} \begin{pmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix},$$

with $K_{\text{tr}}^2 = \frac{1}{3}I$ and $K_{\text{tr}}^{-1} = 3K_{\text{tr}}$. It exists and is nondegenerate because $\mathfrak{sl}(2, \mathbb{C})$ is simple (Cartan criterion), and it is the unique invariant contact direction by Lemma 4.

Proof. The adjoint representation of $\mathfrak{sl}(2, \mathbb{C})$ is the spin-1 representation, so $B(x, y) = \text{Tr}_{3 \times 3}(R_1(x)R_1(y))$ exactly. With $\text{Tr}J_z^2 = 2$, $\text{Tr}J_+J_- = 4$, $\text{Tr}J_{\pm}^2 = 0$: $B(T_0, T_0) = 2$, $B(T_{+1}, T_{-1}) = -\frac{1}{2}\text{Tr}J_+J_- = -2$, $B(T_{\pm 1}, T_{\pm 1}) = 0$. $\square$

Remark 1 (One strengthened principle, two consequences). Under H3^+ , the same simplicity that selects $g = 0$ — and hence exactly three families — guarantees that the seesaw contact direction exists, is unique, and is invertible. In a hypothetical $g = 1$ branch the Majorana contact may be invariant and nondegenerate, but it is not supplied by the adjoint trace. “Exactly three families” and “a nondegenerate Killing contact” are therefore two consequences of the added H3^+ selection, not consequences of invariant bilinearity alone.

VI. WHAT IS NOT CLAIMED

The one-family enumeration and the bound $N_{\text{fam}} \leq 3$ are consequences of H1 and H2 , respectively. Saturation at three and the identification of the selected contact with the Killing form require the additional H3^+ selection axiom. These hypotheses are physical choices, and we do not claim them forced. In particular: no claim is made that a specific grand-unified dynamics realizes the package, no flavor fit or proton-decay phenomenology is asserted here, and the *coefficient* multiplying the Killing contact in any realized Majorana sector is an input — in the companion framework this last statement is itself a theorem (the coefficient is provably underivable from hypotheses of the above kind), and the accompanying phenomenological audits are reported elsewhere. The rigid content of this Letter is narrower: one family forces the half-spinor and its sixteenth state; self-carried covariance bounds the family number by three; and, conditionally on H3^+ , the Cartan criterion saturates the bound and hands the seesaw its unique Killing direction.

VII. MACHINE VERIFICATION

The finite enumeration and matrix identities above are verified by standalone audits (plain `python3/numpy`; the enumeration in exact rational arithmetic): the $-w_0$ involutions for all simple algebras of rank 4–15 and E_6, E_7, E_8, F_4 ; the full dimension- ≤ 16 scan with the $\mathfrak{so}(9)/\mathfrak{so}(15)$ near-misses; the exact $SU(15)/SU(16)$ cubic traces; the exactly vanishing cubic trace tensor of the oscillator-built 16 over all 45^3 triples and its demicube weight system; the hypercharge traces and $k_Y = 5/3$; the one-dimensionality of the invariant-contact solution space; the explicit one-dimensional abelian counterexample $B(e, e) = 1$ with zero Killing form; and the identity $B = 2\sqrt{3} K_{\text{tr}}$. The audit scripts and machine-readable ledgers (68 checks) are available in the public repository github.com/boypatrick/GUT_routeA.

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